

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

24th May 2023

Subject SP8 – General Insurance: Pricing

Time allowed: 3 Hours 15 Minutes (14.45 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
3. *Attempt all questions beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1)

- i) Outline the broad steps for pricing of a stop loss reinsurance contract for Professional Indemnity Insurance for medical practitioners. (3)
- ii) List down the important considerations when pricing a stop loss reinsurance cover. (4)
[7]

Q. 2)

- i) Outline the reasons to use original loss curves in the form of Increased Limit Factors (ILFs) rather than in the form of first loss scales for professional indemnity insurance business. (3)
- ii) Construct an empirical exposure curve table for commercial property insurance portfolio of a risk carrier based on below table which shows ground-up losses and sum-insured data of last 3 years, using increments of 25% of sum insured.

Sum Insured (in INR Crore)	Ground-up Loss amount (in INR Crore)
5.40	4.00
10.00	2.40
10.00	8.00
9.50	7.50
6.00	4.00
5.00	2.50
2.50	1.50
10.00	4.00
3.00	0.70
10.00	7.60

(7)
[10]

Q. 3) You have been provided the following data for the portfolio in consideration:

Year	Number of Policies Sold	Total Losses
2018	65	33,150
2019	75	76,500
2020	80	22,400
2021	90	10,800
2022	95	48,450

The distribution of average losses per unit in year t is $X_t | \theta_t$ which follows a $\text{Gamma}(V_t, V_t / \theta_t)$ distribution with shape parameter equal to V_t and rate parameter equal to V_t / θ_t .

θ_t follows a lognormal distribution with parameters $\mu = 5$ and $\sigma^2 = 1$.

Calculate, showing all workings, the total loss in year 2023 by using the Bühlmann-Straub credibility model for the total losses in the year 2023, assuming that 100 policies will be sold in that year.

[10]

Q. 4) An Indian mono-line motor insurance company currently using four rating areas i.e. East, West, North and South. Pricing Actuary wants to increase the number of rating areas to 20 zonal levels by dividing the country according to population density.

Assess and discuss the Pricing Actuary's suggestion.

[10]

Q. 5)

i) Define gradient boosted model and artificial neural networks. (4)

ii) Discuss the two major disadvantages of machine learning models other than Generalized Linear Models (GLM), namely interpretability and explainability, in the context of insurance pricing. (6)

iii) An insurance company uses Generalized Linear Model to price its Motor insurance portfolio.

Discuss the possible reasons why the insurer may still choose to build machine learning models other than GLM, despite the disadvantages described in part (ii) above.

(6)

[16]

Q. 6) The following is the output from a catastrophe model run on insurer's property insurance portfolio, on gross of reinsurance basis:

(Amounts in INR Crore)

Probability of Non-Exceed	OEP	AEP
99.99%	722.7	736.5
99.95%	528.5	540.1
99.90%	419.7	430.9
99.80%	307.4	318.3
99.60%	203.8	215.2
99.50%	176.7	188.3
99.00%	115.6	126.6
98.00%	78.4	87.1
96.00%	52.8	58.8
95.00%	45.8	50.9
90.00%	26.2	29.1
80.00%	20.0	22.4
70.00%	15.0	16.8
60.00%	12.0	13.4
50.00%	10.0	11.2
AAL	17.0	18.4

There is no inuring reinsurance in place.

i) Define Occurrence Exceedance Probability (OEP), Aggregate Exceedance Probability (AEP), and Average Annual Loss (AAL). (3)

ii) The insurer approached a reinsurer to purchase Catastrophe Excess of Loss (CAT XL) reinsurance. The reinsurer is pricing the CAT XL Reinsurance cover with the following loadings: fixed expenses equal to Rs. 2 Crores, variable expenses equal to 10% of the premium, and risk loading (including provision for profit) equal to 30% of the premium.

Ignore all other expenses, commissions, investment income and any other loadings.

Assume that the attachment point for the treaty is zero and that the Limit is equal to 116 Crores.

Assume that the treaty offers unlimited number of free reinstatements.

Calculate the premium for this reinsurance as per the above requirements. State assumptions and approximations, if any, made. (5)

- iii) The insurer, after receiving the pricing for the scenario mentioned above, requested for a higher limit of 177 Crores.

Discuss the considerations which the reinsurer should make in order to arrive at the premium for the revised limit. (4)

- iv) Discuss how insurers can use catastrophe models in underwriting and risk selection in their property insurance business. (5)
- [17]**

Q. 7) Company X is a relatively new general insurer primarily focused on Group Health and Travel Insurance. They are planning to expand their portfolio by offering Personal Motor Insurance in a different manner compared to the standard traditional motor insurance. In line with this business strategy, the marketing department has come up with a proposal for Pay-as-you-Go policy to cover only Motor Own Damage.

The proposal has been shared with the actuarial department for review. The salient features of the proposal are as follows:

- Drivers who expect to drive less than 30 kms per day on an average are the target customer group.
- The proposal raises an open-ended question with respect to renewals of those policyholders who drive more than 11,000 kms in a policy year.
- Policy is applicable to private & single drivers and restricted to self-drive use only.
- Policyholder pays a policy administration charge at the start of every year.
- During the year, the policyholder opts for own damage risk cover daily as per their need. It can be automated as well, to select coverage in advance for balance policy period.
- The premium needs to be paid via the insurer's app before the start of the day (i.e., previous day before midnight). This is a simple one-click process and premium is deducted directly from the prepaid balance / bank / credit card.
- The daily premium is subject to change on a daily basis depending on the extent of usage and driving style/range.
- The car will be serviced in insurer-authorized garages only or else there will be a penalty/deductible levied.

Assume that the source of the telematics data is through a plug in-device which is self-installed and it comes with a minimal cost. Assume that there is no possibility of data tampering or switching off the device.

Assume that the telematics device measures a number of elements of interest to the insurers: miles driven; time of day; where the vehicle is driven (Global Positioning System or GPS); rapid acceleration; hard braking; hard cornering; air bag deployment etc.

Assume that the policyholders who opt for this product are willing to share the data generated by the device.

- i) List the main rating factors which may be used in pricing a standard or traditional personal motor insurance. (4)
 - ii) State four risk factors that are useful but not normally used in pricing the traditional motor insurance policy.

In addition, state the rating factors that insurers may use as a proxy for those four risk factors. (6)
 - iii) Identify key advantages and disadvantages for the policyholder and the insurance company in this policy vs. standard annual personal motor insurance policy. (10)
 - iv) Describe and discuss the challenges you would face in pricing such a policy. (10)
- [30]**
